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INTEGRATED DRIVING BAND OBTURATOR FOR ARTILLERY

Abstract

A projectile that includes an artillery shell having a casing that defines a surface is provided. The projectile has an obturator that includes an obturator first material layer on the casing surface and an obturator second material layer on the obturator first material layer. The obturator also has an obturator third material layer on the obturator second material layer. The projectile also has a driving band that includes a driving band first material layer on the casing surface and a driving band second material layer on the driving band first material layer. Additionally, the projectile includes a driving band lead-in having a driving band lead-in first material layer on the casing surface and a driving band lead-in second material layer on the driving band lead-in first material layer.

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Background/Summary

TECHNICAL FIELD

[0001] Examples relate to projectiles and more specifically to an obturator for projectiles and a method of forming an obturator onto projectiles.

BACKGROUND

[0002] A projectile can be fired from a weapon with a propellant. Typically, the propellant can be formed from various chemical substances that can rapidly generate gas that causes a pressure differential between a aft position of a barrel of the weapon and a fore position of the barrel. The pressure differential can cause the acceleration of projectile through the barrel and from the weapon. Many propellants use exothermic reactions to generate gases that can increase pressure.

[0003] When projectile is within the barrel of a weapon, in order to prevent pressure loss by the generated gases escaping around the projectile instead of being restricted to behind the projectile and thereby pushing the projectile forward and through the barrel, the projectile can include an obturator. The obturator can function as a gasket that can help minimize the possibility of gas escaping around the projectile. In addition, during firing, rifling within the barrel causes rotation of the projectile. In order to stabilize the projectile during rotation and in flight, the projectile can include a rotating band that engages the rifling.

[0004] Typically, the obturator and the rotating band are separate, individual components. These components are attached to the projectile prior to placing the projectile within the weapon, and typically during the manufacturing process. By virtue of being added to the body of the projectile, the components, they can separate from the projectile during firing. If the rotating band separates it can cause damage to the barrel and destabilize the projectile during firing. If the obturator separates, pressure is lost and therefore range and stability can be diminished.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0005] FIG. 1 illustrates, by way of example, a weapon having projectile and propellant, in accordance with examples.

[0006] FIG. 2 shows an obturator, a driving band, and a driving band lead-in formed on the projectile of FIG. 1, in accordance with examples.

[0007] FIGS. 3 and 4 illustrate the formation of the obturator of FIG. 2, in accordance with examples.

[0008] FIG. 5 illustrates the formation of the driving band of FIG. 2, in accordance with examples.

[0009] FIGS. 6 and 7 show the formation of the driving band lead-in of FIG. 2, in accordance with examples.

[0010] FIG. 8 shows a method for forming a projectile, in accordance with examples.

[0011] FIG. 9 shows the formation of a projectile, in accordance with examples.

DETAILED DESCRIPTION

[0012] The following description and the drawings sufficiently illustrate teachings to enable those skilled in the art to practice them. Other embodiments may incorporate structural, logical, electrical, process, and other changes. Portions and features of some examples may be included in, or substituted for, those of other examples. Teachings set forth in the claims encompass all available equivalents of those claims.

[0013] Examples relate to an obturator, a driving band and a driving band lead-in for a projectile and a method of forming the obturator, the driving band, and driving band lead-in on a casing surface of the projectile. The obturator can function as a seal that can minimize the escaping of gases from an aft end of projectile to a forward end of the projectile during firing of the projectile from a weapon. The obturator can be formed at the aft end and can include a plurality of materials. The obturator can include a first material disposed at the casing surface. The first material can be

formed of a material that can form a strong bond with the casing surface. The first material can transition to a second material that can be softer than the first material. The second material can transition to a third material that can be softer than the second material and can have a lower melting point than the second material. The softer nature of the third material can allow the third material to form a seal against a barrel of the weapon during firing. Each of the first material, the second material, and third material can form an obturator material profile. As used herein, a material profile can relate to a distribution of different materials and how materials within the profile transition from one material to the next.

[0014] The driving band can also include the first material disposed at the casing surface. The first material can transition to the second material. Here, the first material and the second material of the driving band can form a driving band material profile. The driving band material profile can be different from the obturator material profile.

[0015] The driving band lead-in can also include the first material disposed at the casing surface that transitions to the second material. In some examples, the driving band lead-in can include the third material where the second material transitions to the third material. The first material and the second material of the driving band lead-in can form a driving band lead-in material profile that can be different from the obturator material profile and the driving band material profile. When the driving band lead-in includes the third material, the driving band lead-in material profile can be different from the obturator material profile and the driving band material profile. An advantage of examples relates to improved adhesion to a casing. Bands can get ripped off. However, examples address this problem by starting with a high adhesive layer and then a blended transition from metal to plastic.

[0016] Now making reference to FIG. 1, a weapon **100** is shown having a projectile **102** and propellant **104**. The propellant **104** can include a chemical substance that causes a pressure differential between an aft end **106** of the weapon **100** and a forward end **108** of the weapon **100**. The pressure differential can cause the acceleration of the projectile **102** through a barrel **110** and from the weapon **100**.

[0017] The projectile **102** can include an obturator **112** where the obturator **112** can abut a driving band **114**. The driving band **114** can be forward of the obturator **112** where the driving band **114** can be closer to the forward end **108** in comparison to the obturator **112** while the obturator **112** can be closer to the aft end **106** in comparison to the driving band **114**. Each of the obturator **112** and the driving band **114** can be directly disposed on a casing surface **116** of the projectile **102**. The obturator **112** can be disposed aft of the driving band **114** and adjacent the propellant **104**. The driving band **114** can function to assist with the rotation of the projectile **102** during discharge of the projectile **102** from the weapon **100**. When the propellant **106** is activated to create a pressure differential between the weapon aft end **106** and the weapon forward end **108**, the obturator **112** can function to minimize gases escaping around the projectile **102** and into the barrel **110** during discharge of the projectile **102** from the weapon **100**.

[0018] Making reference to FIG. 2, the obturator **112** can extend a distance **200** from the projectile casing surface **116**. Furthermore, the driving band **114** can extend a distance **202** from the projectile casing surface **116** that is less than the distance **200**. A driving band lead-in **204** can also be disposed on the projectile casing surface **116**. The driving band lead-in **204** can abut the driving band **114** and be forward of the driving band **114** where the driving band lead-in **204** can be closer to the forward end **108** in comparison to the driving band **114** and the obturator **112**. The driving band lead-in **204** can function to minimize gases from escaping around the projectile **102** and to the forward end **108** when the propellant **106** is activated.

[0019] Each of the obturator **112**, the driving band **114**, and the driving band lead-in **204** can be formed from different materials. As shown with reference to FIG. 3, the obturator **112** can include a first material layer **300** disposed directly on the projectile casing surface **116**. The obturator first material layer **300** can be formed from a first material having a hardness that allows for robust

bonding with the projectile casing surface **116**. The first material of the obturator first material layer **300** can be such that the obturator first material layer **300** does not separate from the projectile casing surface **116** during discharge from the weapon **100**. The hardness can be in a range of a Mohs hardness of 4.0 to a Mohs hardness of 6.0. An example of the first material can be nickel. Further examples can include tungsten and tantalum, Chromium, Molybdenum, and columbium.

[0020] The obturator **112** can also include a second material layer **302** disposed directly on the obturator first material layer **300**. The obturator second material layer **302** can be formed from a second material having a hardness that is less than the first material of the obturator first material layer **300** and can deform when the projectile **102** is discharged from the weapon **100**. The obturator second material layer **302** can deform in order to minimize damage to the barrel **110** when the projectile **102** is discharged from the weapon **100**. The hardness can be in a range of a Mohs hardness of 1.0 to a Mohs hardness of 3.5. An example of the second material can be copper. Further examples can include polyether ether ketone (PEEK), acrylic, or the like. PEEK has a Shore D hardness of about 85 and Rockwell M of around 100. Acrylic a shore D of about 94 Rockwell M of about 70. In further examples, the copper can be an intermediate layer only.

[0021] Additionally, the obturator **112** can have a third material layer **304** disposed directly on the obturator second material layer **302**. The obturator third material layer **304** can be formed from a third material having a hardness that is less than the first material for the obturator first material layer **300** and the second material for the second obturator material layer **302**. The third obturator material layer **304** can be formed from a material that can deform and form a seal **400** (FIG. 4) between the obturator third material layer **304** and the barrel **110** when the projectile **102** is discharged from the weapon **100**. When the seal **400** is formed, the obturator third material layer can form a gasket that minimizes gases escaping around the projectile **102** and into the barrel **110** during discharge of the projectile **102** from the weapon **100**. An example of the material can be High Density Polyethylene (HDPE). Further examples can include PEEK and acrylic.

[0022] The obturator first material layer **300** can include a transition layer **306** defined between the obturator first material layer **300** and the obturator second material layer **302**. The obturator transition layer **306** can include both the first material and the second material where the obturator transition layer **306** can include a higher concentration of the first material near an area **308** and a higher concentration of the second material near an area **310**. To further illustrate, at the area **308**, the obturator transition layer **306** can have 100% of the first material and at the area **310** the obturator transition layer **306** can have 100% of the second material. The material concentration can vary linearly across the obturator transition layer **306** such that at a midpoint **312** the obturator transition layer **306** can have 50% of the first material and 50% of the second material. However, the material concentration can vary in any manner across the obturator transition layer **306** such that at the midpoint **312**, the obturator transition layer **306** can have 80% of the first material and 20% of the second material, 60% of the first material and 40% of the second material, 40% of the first material and 60% of the second material, or vary in an exponential manner across the transition layer **306**, or the like.

[0023] The obturator second material layer **302** can include a transition layer **314** defined between the obturator second material layer **302** and the obturator third material layer **304**. The obturator transition layer **314** can include both the second material and the third material where the obturator transition layer **314** can include a higher concentration of the second material near an area **316** and a higher concentration of the third material near an area **318**. To further illustrate, at the area **316**, the obturator transition layer **314** can have 100% of the second material and at the area **318** the obturator transition layer **314** can have 100% of the third material. The material concentration can vary linearly across the obturator transition layer **314** such that at a midpoint **320**, the obturator transition layer **314** can have 50% of the second material and 50% of the third material. However, the material concentration can vary in any manner across the obturator transition layer **314** such

that at the midpoint **330**, the obturator transition layer **314** can have 80% of the second material and 20% of the third material, 60% of the second material and 40% of the third material, 40% of the second material and 60% of the third material, or vary in an exponential manner across the transition layer **314**, or the like.

[0024] The obturator **112** can have an obturator material profile **322**. The obturator material profile **322** can be defined by the obturator first material layer **300**, the obturator second material layer **302**, and the obturator third material layer **304**. In addition to being defined by the material layers **300-304**, the obturator material profile **322** can be defined by the obturator transition layers **306** and **314** where the obturator transition layers **306** and **314** can define the obturator material profile **322** in conjunction with the material layers **300-304**.

[0025] As noted above, the driving band **114** can be formed from different materials, as shown with reference to FIG. 5. The driving band **114** can include a first material layer **500** disposed directly on the projectile casing surface **116**. The driving band first material layer **500** can be formed from the first material as discussed above with reference to the obturator first material layer **300**. The driving band **114** can also include a second material layer **502** disposed directly on the driving band first material layer **500**. The driving band second material layer **502** can be formed from the second material as discussed above with reference to the obturator second material layer **302**.

[0026] The driving band first material layer **500** can include a transition layer **504** defined between the driving band first material layer **500** and the driving band second material layer **502**. The driving band transition layer **504** can include both the first material and the second material where the driving band transition layer **504** can include a higher concentration of the first material near an area **506** and a higher concentration of the second material near an area **508**. To further illustrate, at the area **506**, the driving band transition layer **504** can have 100% of the first material and at the area **508** the driving band transition layer **504** can have 100% of the second material. The material concentration can vary linearly across the driving band transition layer **504** such that at a midpoint **510** the driving band transition layer **504** can have 50% of the first material and 50% of the second material. The material concentration can vary in any manner across the driving band transition layer **504** such that at the midpoint **510**, the driving band transition layer **504** can have 80% of the first material and 20% of the second material, 60% of the first material and 40% of the second material, 40% of the first material and 60% of the second material, or vary in an exponential manner across the db transition layer **504**, or the like.

[0027] The driving band **114** can have a driving band material profile **512**. The driving band material profile **512** can be defined by the driving band first material layer **500** and the driving band second material layer **502**. In addition to being defined by the material layers **500** and **502**, the driving band material profile **512** can be defined by the driving band transition layer **504** where the driving band transition layer **504** can define the driving band material profile **512** in conjunction with the material layers **500** and **502**.

[0028] The driving band driving band material profile **512** can differ from the obturator material profile **322**. In particular, the driving band **114** can include more of the first material than the obturator **112** and less of the second material than the obturator **112**. Thus, a width **514** corresponding to an amount of the first material in the driving band first material layer **500** can be greater than a width **516** corresponding to an amount of the first material in the obturator first material layer **300**. As noted above, the driving band **114** can function to assist with the rotation of the projectile **102** during discharge of the projectile **102** from the weapon **100**. Moreover, as noted above, the driving band first material layer **500** can be formed with a material having a high a hardness that allows for robust bonding with the projectile casing surface **116**. During discharge, the driving band **114** can be subjected to high angular accelerations. In the past, high angular acceleration would contribute to the rotating band separating from the projectile during discharge. By virtue of employing the first material having the properties discussed above in the driving band first material layer **500**, separation can be minimized due to the robust bonding.

[0029] The driving band lead-in **204** can also be formed from different materials, as shown with reference to FIG. **6**. The driving band lead-in **204** can include a first material layer **600** disposed directly on the projectile casing surface **116**. The driving band lead-in first material layer **600** can be formed from the first material as discussed above with reference to the obturator first material layer **300**. The driving band lead-in **204** can also include a second material layer **602** disposed directly on the driving band lead-in first material layer **600**. The driving band lead-in second material layer **602** can be formed from the second material as discussed above with reference to the obturator second material layer **302**.

[0030] The driving band lead-in first material layer **600** can include a transition layer **604** defined between the driving band lead-in first material layer **600** and the driving band lead-in second material layer **602**. The driving band lead-in transition layer **604** can include both the first material and the second material where the driving band lead-in transition layer **604** can include a higher concentration of the first material near an area **606** and a higher concentration of the second material near an area **608**. To further illustrate, at the area **606**, the driving band lead-in transition layer **604** can have 100% of the first material and at the area **608** the driving band lead-in transition layer **604** can have 100% of the second material. The material concentration can vary linearly across the driving band lead-in transition layer **604** such that at a midpoint **610** the driving band lead-in transition layer **604** can have 50% of the first material and 50% of the second material. The material concentration can vary in any manner across the driving band lead-in transition layer **604** such that at the midpoint **610**, the driving band lead-in transition layer **604** can have 80% of the first material and 20% of the second material, 60% of the first material and 40% of the second material, 40% of the first material and 60% of the second material, or vary in an exponential manner across the driving band lead-in transition layer **604**, or the like.

[0031] The driving band lead-in **204** can have a driving band lead-in material profile **612**. The driving band lead-in material profile **612** can be defined by the driving band lead-in first material layer **600** and the driving band lead-in second material layer **602**. Besides being defined by the material layers **600** and **602**, the driving band lead-in material profile **612** can be defined by the driving band lead-in transition layer **604** where the driving band lead-in transition layer **604** can define the driving band material profile **612** with the material layers **600** and **602**. Moreover, the driving band lead-in material profile **612** can differ from the obturator material profile **322** and driving band material profile **512**.

[0032] In alternative examples, the driving band lead-in **204** can include a third material layer **700**. The driving band lead-in third material layer **700** can be disposed directly on the driving band lead-in second material layer **602**. The third driving band lead-in third material layer **700** can be formed from the third material as discussed above with reference to the obturator third material layer **304**.

[0033] In alternative examples, the driving band lead-in second material layer **602** can include a transition layer **702** defined between the driving band lead-in second material layer **602** and the driving band lead-in third material layer **700**. The driving band lead-in transition layer **702** can include both the second material and the third material where the driving band lead-in transition layer **702** can include a higher concentration of the second material near an area **704** and a higher concentration of the third material near an area **706**. To further illustrate, at the area **704**, the driving band lead-in transition layer **702** can have 100% of the second material and at the area **706** the driving band lead-in transition layer **702** can have 100% of the third material. The material concentration can vary linearly across the driving band lead-in transition layer **702** such that at a midpoint **708**, the driving band lead-in transition layer **702** can have 50% of the second material and 50% of the third material. However, the material concentration can vary in any manner across the driving band lead-in transition layer **702** such that at the midpoint **708**, the driving band lead-in transition layer **702** can have 80% of the second material and 20% of the third material, 60% of the second material and 40% of the third material, 40% of the second material and 60% of the third material, or vary in an exponential manner across the transition layer **702**, or the like.

[0034] In alternative examples, the driving band lead-in **204** can have a driving band lead-in material profile **710**. The driving band lead-in material profile **710** can be defined by the driving band lead-in first material layer **600**, the driving band lead-in second material layer **602**, and the driving band lead-in third material layer **704**. In addition to being defined by the material layers **600**, **602**, and **704**, the driving band lead-in material profile **710** can be defined by the driving band lead-in transition layers **604** and **702** where the driving band lead-in transition layers **604** and **702** can define the driving band lead-in material profile **710** in conjunction with the material layers **600**, **602**, and **704**. The driving band lead-in material profile **710** can differ from the obturator material profile **322** and the driving band material profile **512**. In particular, the driving band lead-in first material layer **600** can have a width **712**, which can differ from the driving band first material layer width **512** and the obturator first material layer width **516**. Similarly, the driving band lead-in second material layer can vary in width from the obturator second material layer **302** and the driving band second material layer **502**. Moreover, the driving band lead-in third material layer **700** can vary in width from the obturator third material layer **304**.

[0035] As noted above, HDPE, PEEK, and/or acrylic in the obturator third material layer **304** can deform during discharge and form a seal **400** with the barrel **110**. In examples where the driving band lead-in includes the driving band lead-in third material layer **700** that is formed with HDPE, during discharge, the driving band lead-in **204** can also form a gasket similar to the seal **400** towards the forward end **108**. Thus, when a seal is formed to form the gasket, the additional gasket formed by the driving band lead-in third material layer **700** can also minimize gases from escaping around the projectile **102** and into the barrel **110**.

[0036] Now making reference to FIG. **8**, a method **800** for forming features on a projectile in accordance with examples is shown. Initially, first material layers are formed on a casing surface of a projectile. In particular, during an operation **802**, a first material can be directly deposited on the projectile casing surface to form an obturator first material layer. Moreover, during an operation **804**, first material can be directly deposited on the projectile casing surface to form a driving band first material layer. The operation **804** can be performed such that the driving band material layer abuts the obturator first material layer. During an operation **806**, a first material can be directly deposited on the projectile casing surface to form a driving band lead-in first material layer. The operation **806** can be performed such that the driving band lead-in first material layer abuts the driving band first material layer. The first material layer used to form the first material layers in the operations **802-806** can include a first material having a first hardness, such as nickel, as discussed above.

[0037] The operations **802-806** can be implemented with coating deposit method such as a gas dynamic cold spraying procedure. In particular, the first material in the form of a solid powder can be accelerated in a supersonic gas jet at the projectile casing surface. When the first material impacts the projectile casing surface, the first material can undergo plastic deformation and bond with projectile casing surface at the molecular level. This type of bonding, coupled with the use of a material having a hardness in a range of a Mohs hardness of 4.0 to a Mohs hardness of 6.0, can result in robust bonding having the advantages discussed above.

[0038] Moreover, during a cold spraying process, the operations **802-806** can be performed at the same time. In particular, a nozzle can move along direction X as shown in FIG. **9** and deposit first material for the obturator first material layer **300** at an obturator area **900**, first material for the driving band first material layer **300** at a driving band area **902**, and first material for the driving band lead-in first material layer **300** at a driving band lead-in area **904**. Moreover, during the operations **802-806** and the cold-spraying process, the projectile **102** can be rotated along either direction A or direction B and the nozzle can move along direction Y as shown in FIG. **9** and deposit first material for the obturator first material layer **300**, the driving band first material layer **300**, and the driving band lead-in first material layer **300**. As discussed herein, the projectile **102** can be rotated along the directions A or B once the nozzle completes a pass along either the

directions X or Y.

[0039] In the method **800**, during an operation **808**, a second material having a second hardness that is less than the first hardness can be deposited directly on the first obturator first material layer. The second material can be deposited to form an obturator second material layer. The obturator second material layer can be formed with a second material such as copper as discussed above. The obturator second material layer can be deposited with a cold-spraying process as discussed above.

[0040] As noted above, the obturator material profile **322**, the driving band material profile **512**, and the driving band lead-in material profile **710** can vary from each other based on the differing amounts of material in each respective layer, such as the first material layers for each of the obturator, the driving band, and the driving band lead-in. Thus, when the operation **808** is being performed to form the obturator second material layer, the operation **804** can continue to be performed to form the driving band first material layer using a cold spray process where the first material is fed to the nozzle while the nozzle passes over the driving band area **902**.

[0041] The method **800** can perform an operation **810** simultaneously with the operation **808** or when the operation **808** is complete. In the operation **810**, the second material can be directly deposited on the driving band first material layer to form a second driving band material layer. A portion of the driving band first material layer can abut a portion of the obturator first material layer while another portion of the driving band first material layer can abut the obturator second material layer. Moreover, a portion of the deposited driving band second material layer can abut a portion the obturator second material layer. The driving band first material layer formed during the operation **804** and the driving band second material layer formed during the operation **810** can form a driving band material profile, such as the driving band material profile **512** having the characteristics described above. The operation **810** can be performed with a cold spray process as discussed above.

[0042] The method **800** can also perform an operation **812**, where a second material can be directly deposited on the driving band lead-in first material layer to form a driving band lead-in second material layer. The operation **812** can be performed simultaneously with the operations **808** and **810** or after one or both of the operations **808** and **810**, or a combination thereof. Additionally, the operation **812** can be performed with a cold spray process as discussed above. The driving band lead-in first material layer and the driving band lead-in second material layer can form a driving band lead-in material profile, such as the driving band lead-in material profile **612** having the characteristics described above.

[0043] Moreover, the method **800** can perform an operation **814**, where a third material having a third hardness less than the second hardness can be deposited directly on the obturator second material layer to form an obturator third material layer. An example of the third material can be HDPE as described above. The operation can be performed simultaneously with the operations **804** and **812**, simultaneously with the operations **810** and **812**, or after the operations **804**, **810**, and **812**, or a combination thereof. Furthermore, the operation **814** can be performed with a cold spray process as discussed above.

[0044] In examples, the obturator first material layer, the obturator second material layer, and the obturator third material layer can form an obturator material profile, such as the obturator material profile **322** having the characteristics described above.

[0045] In an alternative example, in addition to performing the operations **802-814**, a further operation can be performed where a third material having a third hardness less than the second hardness is deposited directly on the driving band lead-in second material layer. Again, the third material can be HDPE. The third material can be deposited on the driving band lead-in second material layer simultaneously with the operations **804** and **814**, the operations **810** and **814**, or after the operations **804**, **810**, and **814**, or a combination thereof. Here, the driving band lead-in first material layer, the driving band lead-in second material layer, and the third driving band lead-in material layer can form a driving band lead-in material profile, such as the driving band lead-in

material profile 710 having the characteristics described above.

ADDITIONAL EXAMPLES

[0046] Example 1 is a projectile comprising: an artillery shell having a casing that defines a surface; an obturator disposed at the casing surface, the obturator including: an obturator first material layer disposed directly on the casing surface, the obturator first material layer being formed of a first material having a first hardness; an obturator second material layer disposed directly on the obturator first material layer, the obturator second material layer being formed of a second material having a second hardness that is less than the first hardness; and an obturator third material layer disposed directly on the obturator second material layer, the obturator third material layer being formed of a third material having a third hardness that is less than the second hardness; a driving band disposed on the casing layer abutting the obturator and forward the obturator, the driving band including: a driving band first material layer disposed directly on the casing surface and formed of the first material; and a driving band second material layer disposed directly on the driving band first material layer and formed of the second material; and a driving band lead-in disposed on the casing layer abutting the driving band and forward the driving band, the driving band lead-in including: a driving band lead-in first material layer disposed directly on the casing surface and formed of the first material; and a driving band lead-in second material layer disposed directly on the driving band lead-in first material layer and formed of the second material.

[0047] In Example 2, the subject matter of Example 1 includes, wherein: the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator material profile; the driving band first material layer and the driving band second material layer form a driving band material profile that is different from the obturator material profile; the driving band lead-in first material layer and the driving band lead-in second material layer form a driving band lead-in material profile that is different from the obturator material profile and driving band material profile; and the driving band lead-in further comprises a driving band lead-in third material layer disposed directly on the driving band lead-in second material layer, the driving band lead-in third material layer being formed of the third material where the driving band lead-in first material layer, the driving band lead-in second material layer, and the driving band lead-in third material layer form the driving band lead-in material profile.

[0048] In Example 3, the subject matter of Example 2 includes, wherein each of the obturator first, second, and third material layers, each of the driving band first and second material layers, and each of the driving band lead-in first, second, and third material layers are cold-sprayed.

[0049] In Example 4, the subject matter of Examples 2-3 includes, wherein the first material is nickel, the second material is copper, and the third material is one of High Density Polyethylene, Polyether ether ketone, or acrylic.

[0050] In Example 5, the subject matter of Examples 2-4 includes, wherein an obturator transition layer is defined between the obturator first material layer and the obturator second material layer and a driving band transition layer is defined between the driving band first material layer and the driving band second material layer where a composition of the obturator transition layer is different from a composition of the driving band transition layer.

[0051] In Example 6, the subject matter of Example 5 includes, wherein a driving band lead-in transition layer is defined between the driving band lead-in first material layer and the driving band lead-in second material layer, where a composition of the driving band lead-in layer transition layer is different from the obturator transition layer composition and the driving band transition layer composition.

[0052] In Example 7, the subject matter of Examples 1-6 includes, wherein the obturator extends a first distance away from the casing surface and the driving band extends a second distance away from the casing surface where the first distance is greater than the second distance.

[0053] Example 8 is a method of forming a projectile having an artillery shell with a casing that defines a surface, the method comprising: depositing a first material having a first hardness directly

on the casing surface to form an obturator first material layer; depositing the first material directly on the casing surface and abutting the obturator first material layer to form a driving band first material layer; depositing the first material directly on the casing surface abutting the driving band first material layer to form a driving band lead-in first material layer; depositing a second material having a second hardness that is less than the first hardness directly on the obturator first material layer to form an obturator second material layer; depositing the second material directly on the driving band first material layer to form a driving band second material layer where a portion of the driving band first material layer abuts the obturator first material layer and the obturator second material layer and a portion of the driving band second material layer abuts the obturator second material layer, wherein the driving band first material layer and the driving band second material layer form a driving band material profile; depositing the second material directly on the driving band lead-in first material layer to form a driving band lead-in second material layer, wherein the driving band lead-in first material layer and the driving band lead-in second material layer form a driving band lead-in material profile; and depositing a third material having a third hardness that is less than the second hardness directly on the obturator second material layer to form an obturator third material layer, wherein the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator material profile where the driving band material profile is different from the obturator material profile and the driving band lead-in material profile is different from the driving band material profile and the obturator material profile.

[0054] In Example 9, the subject matter of Example 8 includes, wherein each of the depositing operations are cold spray depositing operations.

[0055] In Example 10, the subject matter of Examples 8-9 includes, the method further comprising depositing the third material directly on the driving band lead-in second material layer to form a driving band lead-in third material layer where the driving band lead-in first material layer, the driving band lead-in second material layer, and the driving band lead-in third material layer form the driving band lead-in material profile.

[0056] In Example 11, the subject matter of Example 10 includes, wherein the first material is nickel, the second material is copper, and the third material is one of High Density Polyethylene., Polyether ether ketone, or acrylic.

[0057] In Example 12, the subject matter of Examples 10-11 includes, the method further comprising: depositing an obturator transition layer is between the obturator first material layer and the obturator second material layer; depositing a driving band transition layer between the driving band first material layer and the driving band second material layer where a composition of the obturator transition layer is different from a composition of the driving band transition layer; and depositing a driving band lead-in transition layer between the driving band lead-in first material layer and the driving band lead-in second material layer, where a composition of the driving band layer transition layer is different from the obturator transition layer composition and the driving band transition layer composition.

[0058] In Example 13, the subject matter of Examples 8-12 includes, wherein the obturator extends a first distance away from the casing surface and the driving band extends a second distance away from the casing surface where the first distance is greater than the second distance.

[0059] Example 14 is a projectile comprising: an artillery shell having a casing that defines a surface; an obturator first material layer disposed directly on the casing surface; an obturator second material layer disposed directly on the obturator first material layer; an obturator third material layer disposed directly on the obturator second material layer; a driving band first material layer disposed directly on the casing surface; and a driving band second material layer disposed directly on the driving band first material layer; a driving band lead-in first material layer disposed directly on the casing surface; and a driving band lead-in second material layer disposed directly on the driving band lead-in first material layer.

[0060] In Example 15, the subject matter of Example 14 includes, wherein: the obturator first

material layer is formed of a first material having a first hardness, the obturator second material layer is formed of a second material having a second hardness that is less than the first hardness, the obturator third material layer is formed of a third material having a third hardness that is less than the second hardness, wherein the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator material profile; the driving band first material layer being formed of the first material and the driving band second material layer being formed of the second material where the driving band first material layer and the driving band second material layer form a driving band material profile that is different from the obturator material profile; and the driving band lead-in first material layer being formed of the first material and the driving band lead-in second material layer being formed of the second material, wherein the driving band lead-in first material layer and the driving band lead-in second material layer form a driving band lead-in material profile that is different from the obturator material profile and driving band material profile.

[0061] In Example 16, the subject matter of Example 15 includes, wherein the driving band lead-in further comprises a driving band lead-in third material layer disposed directly on the driving band lead-in second material layer, the driving band lead-in third material layer being formed of the third material where the driving band lead-in first material layer, the driving band lead-in second material layer, and the driving band lead-in third material layer form the driving band lead-in material profile.

[0062] In Example 17, the subject matter of Example 16 includes, wherein each of the obturator material layers, each of the driving band material layers, and each of the driving band lead-in materials layers are cold-sprayed during the same fabrication process.

[0063] In Example 18, the subject matter of Examples 16-17 includes, wherein an obturator transition layer is defined between the obturator first material layer and the obturator second material layer and a driving band transition layer is defined between the driving band first material layer and the driving band second material layer where a composition of the obturator transition layer is different from a composition of the driving band transition layer.

[0064] In Example 19, the subject matter of Example 18 includes, wherein a driving band lead-in transition layer is defined between the driving band lead-in first material layer and the driving band lead-in second material layer, where a composition of the driving band layer transition layer is different from the obturator transition layer composition and the driving band transition layer composition.

[0065] In Example 20, the subject matter of Examples 14-19 includes, wherein the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator that extends a first distance away from the casing surface and the driving band first material layer and the driving band second material layer form a driving band that extends a second distance away from the casing surface where the first distance is greater than the second distance.

[0066] Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement of any of Examples 1-20.

[0067] Example 22 is an apparatus comprising means to implement of any of Examples 1-20.

[0068] Although teachings have been described with reference to specific example teachings, it will be evident that various modifications and changes may be made to these teachings without departing from the broader spirit and scope of the teachings. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. The accompanying drawings that form a part hereof, show by way of illustration, and not of limitation, specific teachings in which the subject matter may be practiced. The teachings illustrated are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed herein. Other teachings may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. This Detailed

Description, therefore, is not to be taken in a limiting sense, and the scope of various teachings is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

Claims

1. A projectile comprising: an artillery shell having a casing that defines a surface; an obturator disposed at the casing surface, the obturator including: an obturator first material layer disposed directly on the casing surface, the obturator first material layer being formed of a first material having a first hardness; an obturator second material layer disposed directly on the obturator first material layer, the obturator second material layer being formed of a second material having a second hardness that is less than the first hardness; and an obturator third material layer disposed directly on the obturator second material layer, the obturator third material layer being formed of a third material having a third hardness that is less than the second hardness; a driving band disposed on the casing layer abutting the obturator and forward the obturator, the driving band including: a driving band first material layer disposed directly on the casing surface and formed of the first material; and a driving band second material layer disposed directly on the driving band first material layer and formed of the second material; and a driving band lead-in disposed on the casing layer abutting the driving band and forward the driving band, the driving band lead-in including: a driving band lead-in first material layer disposed directly on the casing surface and formed of the first material; and a driving band lead-in second material layer disposed directly on the driving band lead-in first material layer and formed of the second material.
2. The projectile of claim 1, wherein: the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator material profile; the driving band first material layer and the driving band second material layer form a driving band material profile that is different from the obturator material profile; the driving band lead-in first material layer and the driving band lead-in second material layer form a driving band lead-in material profile that is different from the obturator material profile and driving band material profile; and the driving band lead-in further comprises a driving band lead-in third material layer disposed directly on the driving band lead-in second material layer, the driving band lead-in third material layer being formed of the third material where the driving band lead-in first material layer, the driving band lead-in second material layer, and the driving band lead-in third material layer form the driving band lead-in material profile.
3. The projectile of claim 2, wherein each of the obturator first, second, and third material layers, each of the driving band first and second material layers, and each of the driving band lead-in first, second, and third material layers are cold-sprayed.
4. The projectile of claim 2, wherein the first material is nickel, the second material is copper, and the third material is one of High Density Polyethylene., Polyether ether ketone, or acrylic.
5. The projectile of claim 2, wherein an obturator transition layer is defined between the obturator first material layer and the obturator second material layer and a driving band transition layer is defined between the driving band first material layer and the driving band second material layer where a composition of the obturator transition layer is different from a composition of the driving band transition layer.
6. The projectile of claim 5, wherein a driving band lead-in transition layer is defined between the driving band lead-in first material layer and the driving band lead-in second material layer, where a composition of the driving band lead-in layer transition layer is different from the obturator transition layer composition and the driving band transition layer composition.
7. The projectile of claim 1, wherein the obturator extends a first distance away from the casing surface and the driving band extends a second distance away from the casing surface where the first distance is greater than the second distance.

8. A method of forming a projectile having an artillery shell with a casing that defines a surface, the method comprising: depositing a first material having a first hardness directly on the casing surface to form an obturator first material layer; depositing the first material directly on the casing surface and abutting the obturator first material layer to form a driving band first material layer; depositing the first material directly on the casing surface abutting the driving band first material layer to form a driving band lead-in first material layer; depositing a second material having a second hardness that is less than the first hardness directly on the obturator first material layer to form an obturator second material layer; depositing the second material directly on the driving band first material layer to form a driving band second material layer where a portion of the driving band first material layer abuts the obturator first material layer and the obturator second material layer and a portion of the driving band second material layer abuts the obturator second material layer, wherein the driving band first material layer and the driving band second material layer form a driving band material profile; depositing the second material directly on the driving band lead-in first material layer to form a driving band lead-in second material layer, wherein the driving band lead-in first material layer and the driving band lead-in second material layer form a driving band lead-in material profile; and depositing a third material having a third hardness that is less than the second hardness directly on the obturator second material layer to form an obturator third material layer, wherein the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator material profile where the driving band material profile is different from the obturator material profile and the driving band lead-in material profile is different from the driving band material profile and the obturator material profile.

9. The method of claim 8, wherein each of the depositing operations are cold spray depositing operations.

10. The method of claim 8, the method further comprising depositing the third material directly on the driving band lead-in second material layer to form a driving band lead-in third material layer where the driving band lead-in first material layer, the driving band lead-in second material layer, and the driving band lead-in third material layer form the driving band lead-in material profile.

11. The method of claim 10, wherein the first material is nickel, the second material is copper, and the third material is one of High Density Polyethylene., Polyether ether ketone, or acrylic.

12. The method of claim 10, the method further comprising: depositing an obturator transition layer is between the obturator first material layer and the obturator second material layer; depositing a driving band transition layer between the driving band first material layer and the driving band second material layer where a composition of the obturator transition layer is different from a composition of the driving band transition layer; and depositing a driving band lead-in transition layer between the driving band lead-in first material layer and the driving band lead-in second material layer, where a composition of the driving band layer transition layer is different from the obturator transition layer composition and the driving band transition layer composition.

13. The method of claim 8, wherein the obturator extends a first distance away from the casing surface and the driving band extends a second distance away from the casing surface where the first distance is greater than the second distance.

14. A projectile comprising: an artillery shell having a casing that defines a surface; an obturator first material layer disposed directly on the casing surface; an obturator second material layer disposed directly on the obturator first material layer; an obturator third material layer disposed directly on the obturator second material layer; a driving band first material layer disposed directly on the casing surface; and a driving band second material layer disposed directly on the driving band first material layer; a driving band lead-in first material layer disposed directly on the casing surface; and a driving band lead-in second material layer disposed directly on the driving band lead-in first material layer.

15. The projectile of claim 14, wherein: the obturator first material layer is formed of a first material having a first hardness, the obturator second material layer is formed of a second material

having a second hardness that is less than the first hardness, the obturator third material layer is formed of a third material having a third hardness that is less than the second hardness, wherein the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator material profile; the driving band first material layer being formed of the first material and the driving band second material layer being formed of the second material where the driving band first material layer and the driving band second material layer form a driving band material profile that is different from the obturator material profile; and the driving band lead-in first material layer being formed of the first material and the driving band lead-in second material layer being formed of the second material, wherein the driving band lead-in first material layer and the driving band lead-in second material layer form a driving band lead-in material profile that is different from the obturator material profile and driving band material profile.

16. The projectile of claim 15, wherein the driving band lead-in further comprises a driving band lead-in third material layer disposed directly on the driving band lead-in second material layer, the driving band lead-in third material layer being formed of the third material where the driving band lead-in first material layer, the driving band lead-in second material layer, and the driving band lead-in third material layer form the driving band lead-in material profile.

17. The projectile of claim 16, wherein each of the obturator material layers, each of the driving band material layers, and each of the driving band lead-in materials layers are cold-sprayed during the same fabrication process.

18. The projectile of claim 16, wherein an obturator transition layer is defined between the obturator first material layer and the obturator second material layer and a driving band transition layer is defined between the driving band first material layer and the driving band second material layer where a composition of the obturator transition layer is different from a composition of the driving band transition layer.

19. The projectile of claim 18, wherein a driving band lead-in transition layer is defined between the driving band lead-in first material layer and the driving band lead-in second material layer, where a composition of the driving band layer transition layer is different from the obturator transition layer composition and the driving band transition layer composition.

20. The projectile of claim 14, wherein the obturator first material layer, the obturator second material layer, and the obturator third material layer form an obturator that extends a first distance away from the casing surface and the driving band first material layer and the driving band second material layer form a driving band that extends a second distance away from the casing surface where the first distance is greater than the second distance.
